Unit 1

1. What are Machine Learning Basics? Explain supervised and unsupervised learning.
2. Explain Capacity, Overfitting and Underfitting with a clear diagram.
3. What are Hyperparameters and what is a Validation Set?
4. What are Estimators? Explain Bias and Variance.
5. Explain Maximum Likelihood Estimation.
6. Explain Bayesian Statistics.
7. What is Stochastic Gradient Descent?
8. Explain the steps involved in Building a Machine Learning Algorithm.
9. What are Deep Feedforward Networks?
10. What is Gradient-Based Learning?
11. What is Backpropagation? Explain with a neat diagram.

1. What are Machine Learning Basics? Explain supervised and unsupervised learning.

1. **Machine Learning (ML)** means computers learn patterns automatically from data.
2. ML improves performance when more data is given.
3. ML is used in prediction, classification, recommendation, image recognition etc.

- **Supervised Learning:**

1. Data contains input + correct output (labels).
2. Model learns by comparing predicted answer with real answer.
3. Used for classification (spam/not spam) and regression (price prediction).

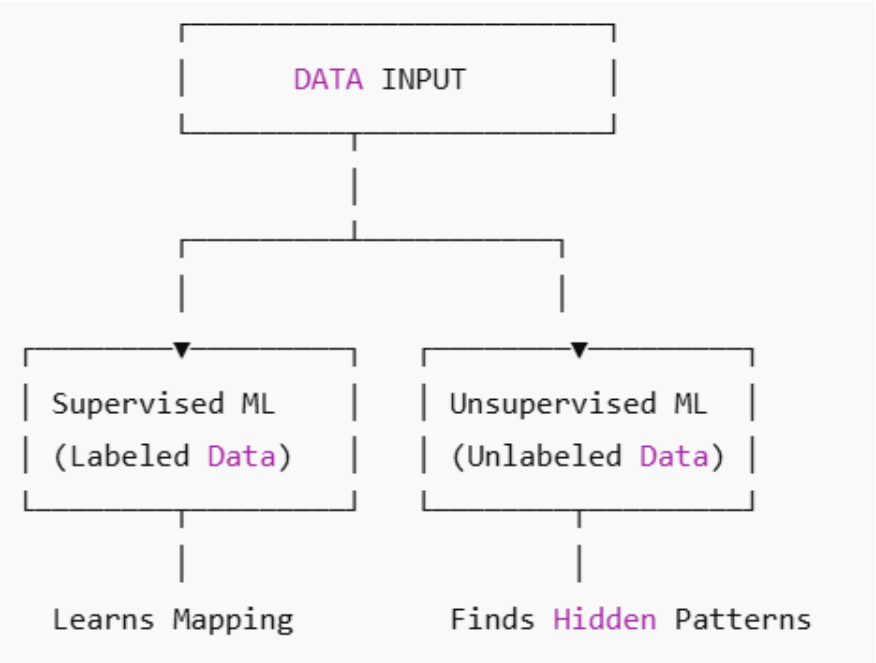
- Examples: Email spam detection, disease prediction.
  Supervised learning is divided mainly into **two types**:
  1) Classification
  Used when the output is a **category** or **label**.

1. Examples:
   – Email spam detection
   – Disease detection (positive/negative)
   – Handwritten digit recognition

Supervised Learning

2) Regression
Used when the output is a **continuous numeric value**.

Examples:
– House price prediction
– Temperature prediction
– Stock market value prediction



Unsupervised Learning:

1. Data has no labels (only input).
2. Model groups or discovers hidden patterns.
3. Useful when we don't know categories.
4. Examples: customer grouping, finding similar images.

In tabular form:

Supervised Learning	Unsupervised Learning
Data has **labels** (answers are known).	Data has **no labels** (answers are unknown).
Model learns using **input + correct output**.	Model learns by **finding patterns on its own**.
Goal: **Predict outputs** for new data.	Goal: **Discover hidden structures** in data.
Used for **classification and regression**.	Used for **clustering and dimensionality reduction**.
Training is more **complex** because labeled data is needed.	Training is **easy** because unlabeled data is sufficient.
High accuracy because of correct answers during training.	Lower accuracy since model learns by itself.
Examples: Spam vs Not Spam, Price prediction.	Examples: Grouping customers, Market segmentation.
Algorithms: Linear Regression, SVM, Decision Trees.	Algorithms: K-Means, PCA, Hierarchical Clustering.
Needs **large labeled datasets**, which is costly.	Works with **unlabeled datasets**, which are cheap.
Output is **known and measurable**.	Output is **unknown and discovered** by the model.

2. Explain Capacity, Overfitting and Underfitting with a clear diagram

- **Model Capacity:**

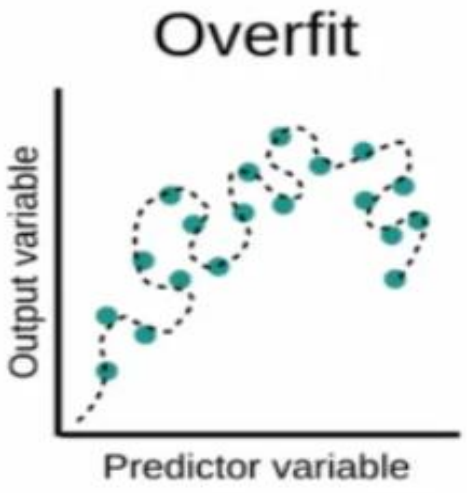
1. Capacity = how well a model can learn patterns.
2. Low capacity model learns few patterns.
3. High capacity model learns complex patterns.
4. Choosing correct capacity is very important.

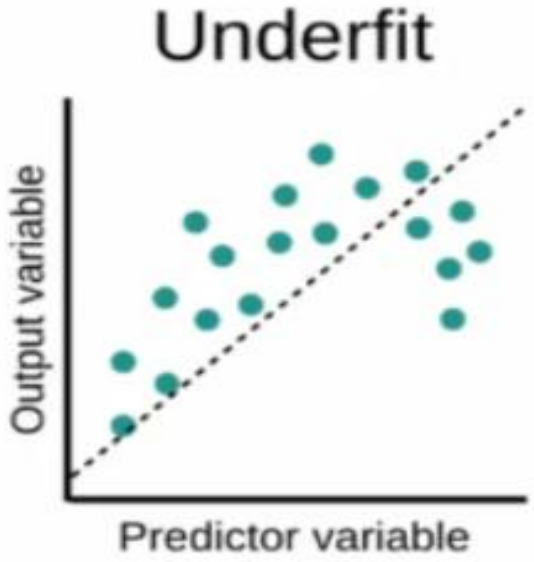
- **Overfitting:**

1. Model memorizes training data.
2. Performs well on training but poorly on new data.
3. Happens due to too many parameters / high capacity.
4. Example: Student memorizing answers but failing new questions.

- **Underfitting:**

1. Model cannot learn basic patterns.
2. Performs poorly on both training and test data.
3. Happens due to too simple model / low capacity.
4. Example: Student not studying enough.





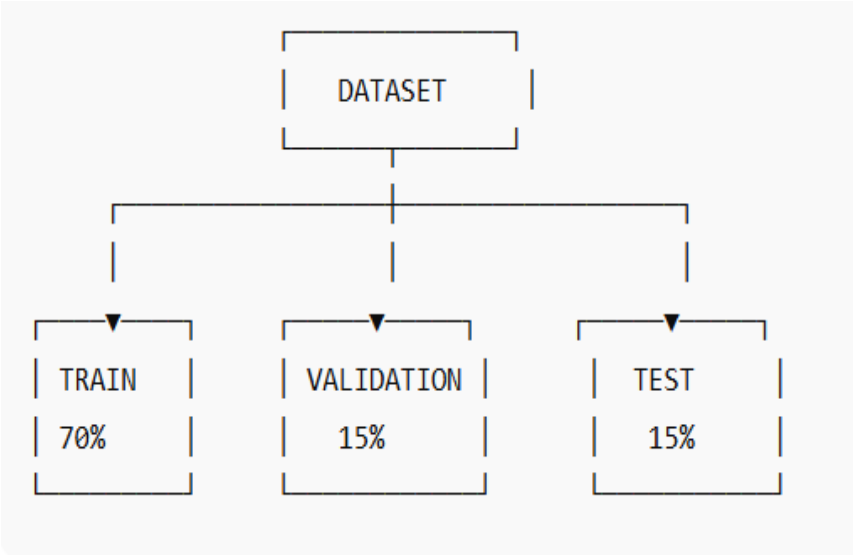
3. What are Hyperparameters and what is a Validation Set?

- **Hyperparameters:**

1. Values set before training (learning rate, layers, batch size).
2. They control learning speed and model behavior.
3. Not learned from data – manually chosen.

- **Validation Set:**

1. Small part of data used to tune hyperparameters.
2. Helps check performance during training.
3. Prevents overfitting and helps select best model.



→The full **dataset** is divided into 3 parts to train and check the model properly.

1. Training Set – 70%

1. Used to **train/teach** the model.
2. Model learns patterns from this data.

2. Validation Set – 15%

1. Used to **tune hyperparameters** (learning rate, layers, etc.).
2. Checks how well the model is performing during training.
3. Helps avoid **overfitting**.

3. Test Set – 15%

1. Used after training is completed.
2. Gives the **final accuracy** of the model.
3. Tells how well the model works on completely new data

Training = Learn

Validation = Improve

Test = Final performance check

4. What are Estimators? Explain Bias and Variance.

- **Estimators:**

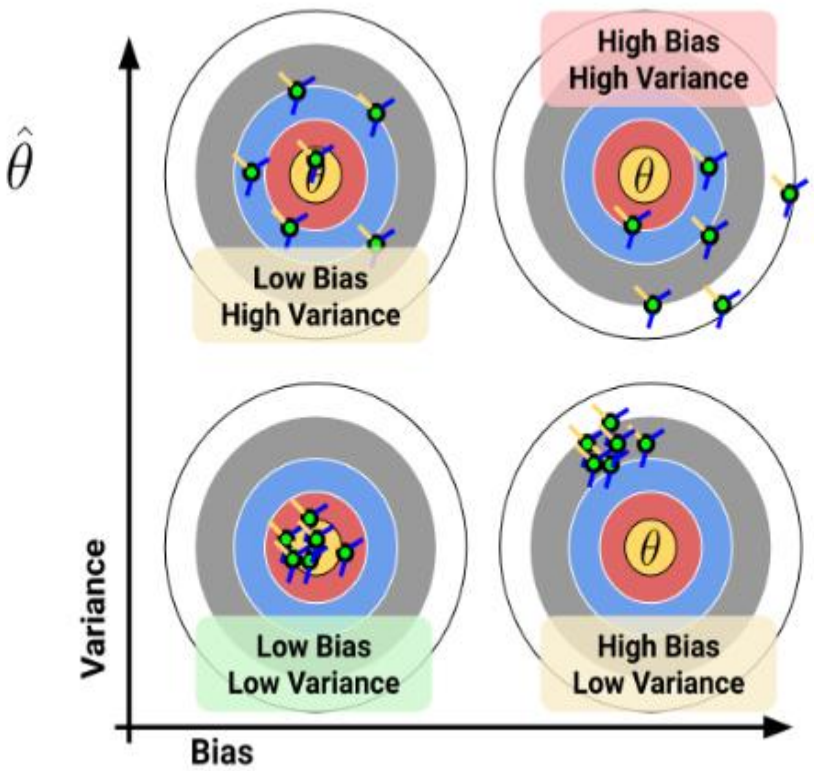
1. Algorithms that estimate values from data (mean, weights, parameters).
2. Used in regression, probability models etc.

- **Bias:**

1. Error caused by oversimplifying the model.
2. High bias → underfitting.
3. Model misses important patterns.

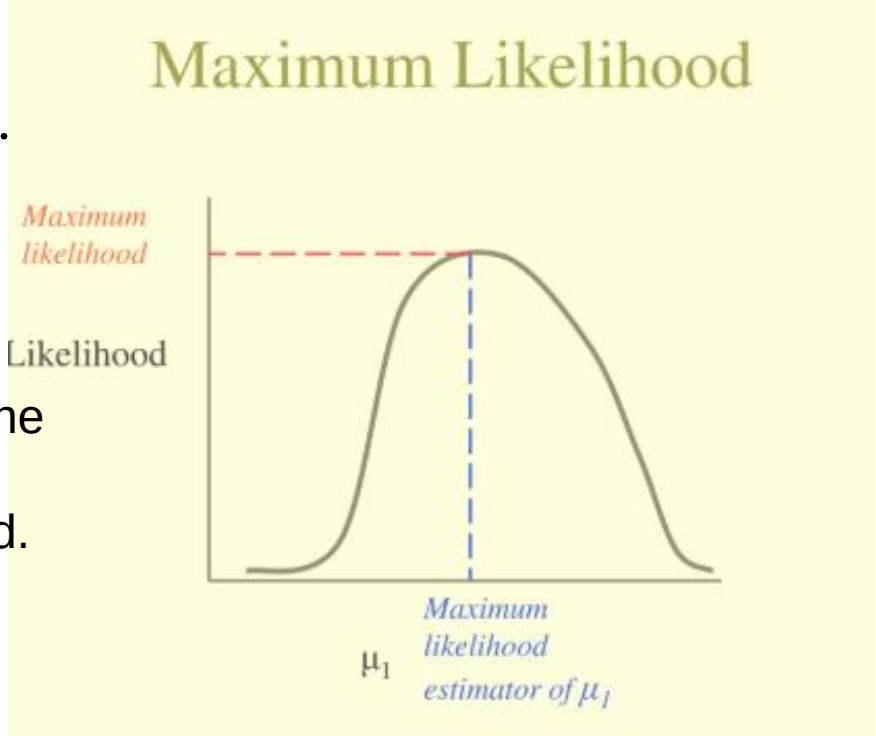
- **Variance:**

1. Error caused by high sensitivity to data noise.
2. High variance → overfitting.
3. Model memorizes data.



5. Explain Maximum Likelihood Estimation.

1. MLE finds parameters that best explain the given data.
2. It selects parameter values that make observed data most probable.
3. Used in regression, classification and probability models.
4. Example: estimating mean of normal distribution.



•**Purpose**: Estimate parameters that best explain the observed data.
•**Idea**: Pick parameter values that **maximize the likelihood** of the data.
•**Likelihood function**: L(θ) = P(data | θ), often use log-likelihood.

•**Steps**:
1.Write L(θ)
2.Take ln L(θ)
3.Differentiate w.r.t θ, set = 0
4.Solve for θ → MLE
•**Properties**: consistent, asymptotically normal, efficient
•**Example**: Coin toss n times, x heads → MLE of p = x/n

6. Explain Bayesian Statistics.

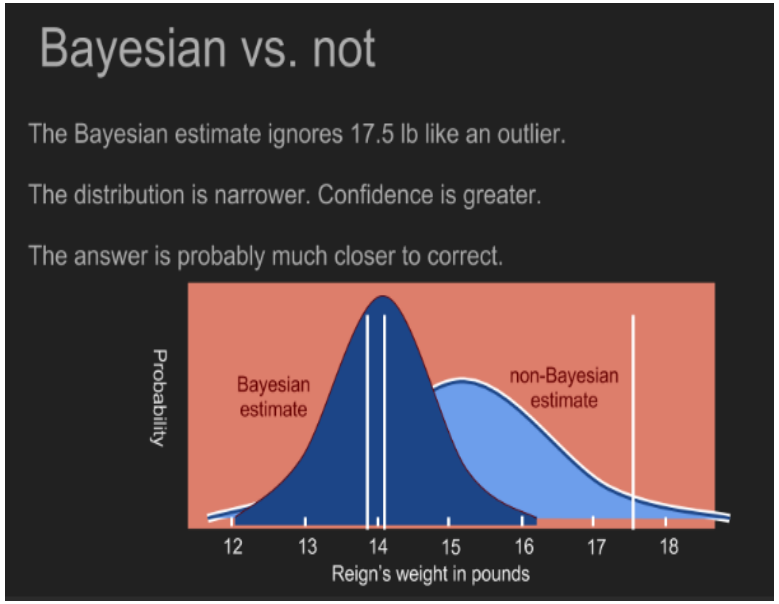
1. Bayesian method updates belief when new information comes.
2. Uses Bayes' theorem: Prior → evidence → posterior.
3. Helpful when data is uncertain or limited.
4. Used in spam filters, diagnosis, risk prediction.

•**Purpose**: Update beliefs about a parameter as new data becomes available.
•**Idea**: Combines **prior knowledge** with **observed data** to make inferences.
•**Bayes' Theorem**:
Posterior ∝ Likelihood × Prior
$P(\theta \mid data) = [P(data \mid \theta) \times P(\theta)] / P(data)$
•**Steps**:

- Choose a **prior** distribution for the parameter.
- Collect **data** and calculate the likelihood.
- Compute **posterior** distribution using Bayes' theorem.
- Make predictions or decisions based on the posterior.

•**Properties**:

- Incorporates prior knowledge
- Provides **full probability distribution** for parameters
- Naturally updates beliefs with new data

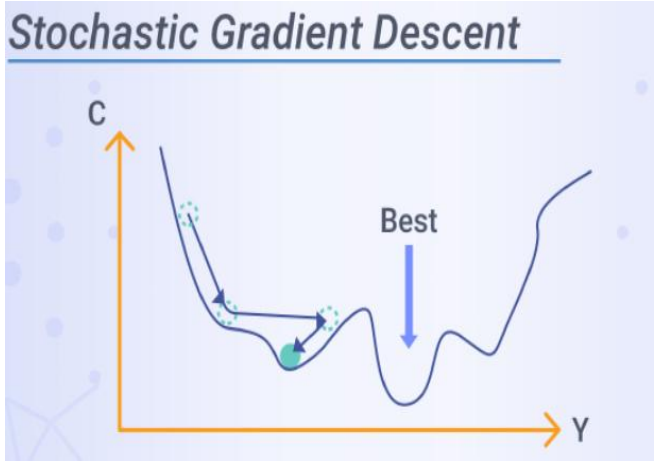


•**Example**: Estimating probability of rain using historical data (prior) and today's weather (data) → updated probability (posterior).

7. What is Stochastic Gradient Descent?

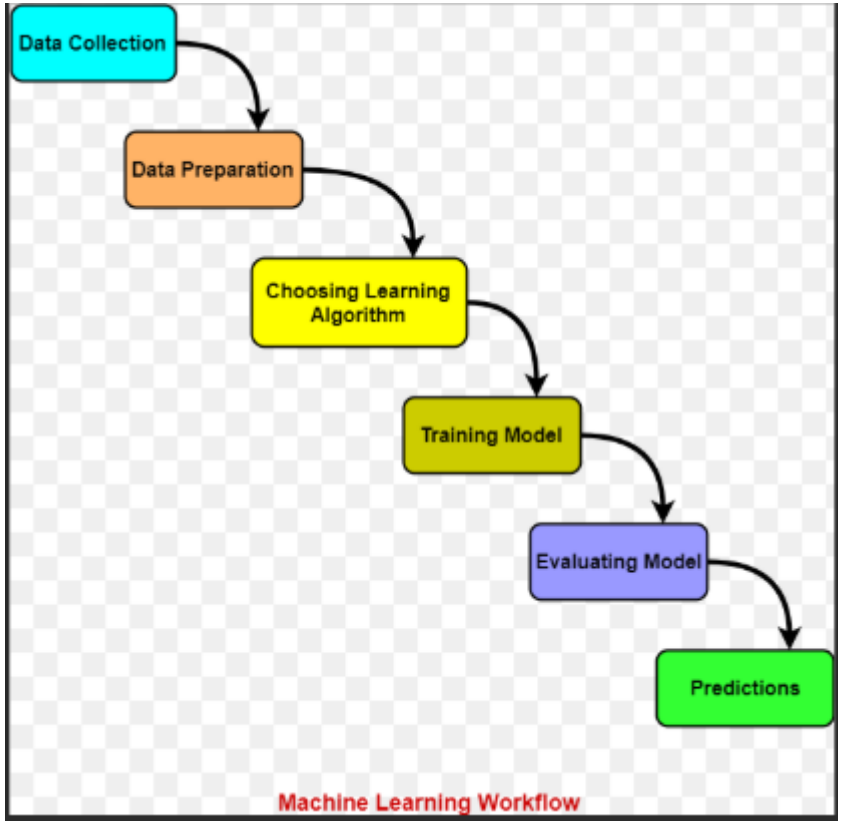
1. SGD updates model using one sample or small batch.
2. Faster than normal gradient descent.
3. Used for large datasets and deep learning.
4. Helps reduce loss step-by-step.

•**Purpose**: Optimize model parameters to **minimize a loss function**.
•**Idea**: Update parameters **step by step using small batches or single data points**, rather than the whole dataset.
•**Key Steps**:
1.Initialize parameters randomly.
2.Pick **one data point (or small batch)**.
3.Compute **gradient** of loss w.r.t parameters.
4.Update parameters: θ = θ − learning_rate × gradient
5.Repeat for many iterations/epochs.
•**Advantages**:
•Faster than full batch gradient descent
•Can escape local minima due to randomness
•**Example**: Training a neural network on large dataset using one sample at a time.



8. Explain the steps involved in Building a Machine Learning Algorithm.

- Collect data.
- Clean and preprocess data.
- Select features.
- Choose model.
- Train model.
- Validate using validation set.
- Test with unseen data.
- Deploy and monitor.



9. What are Deep Feedforward Networks?

- **Deep Forward Networks:**

1. Also called Feedforward Neural Networks.
2. Data moves from input → hidden → output layer.
3. Used for complex pattern learning.

- **Learning XOR:**

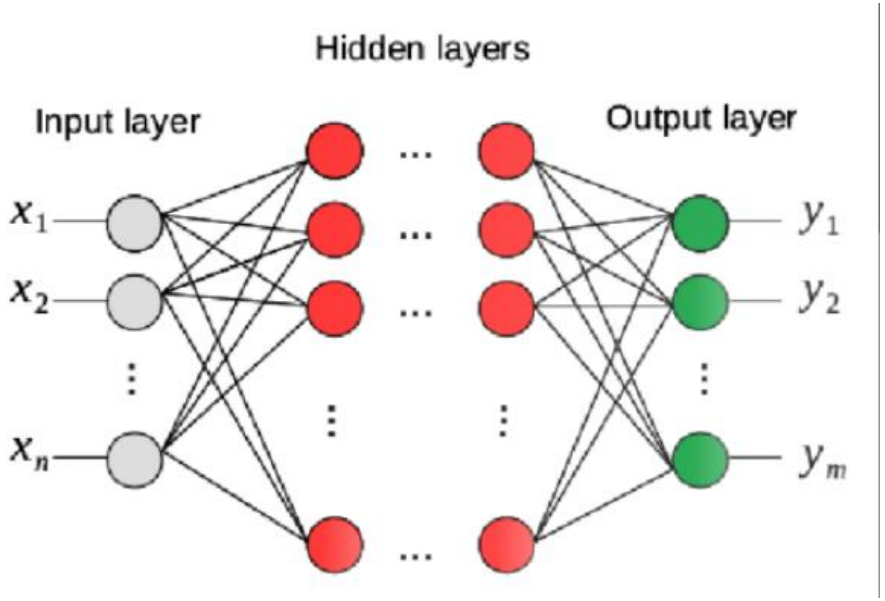
1. XOR cannot be solved by a single-layer network.
2. Requires at least one hidden layer.
3. Proves importance of deep learning.

- **Hidden Units:**

1. Neurons inside hidden layers.
2. They extract features (edges, shapes, patterns).
3. More units → more learning ability.

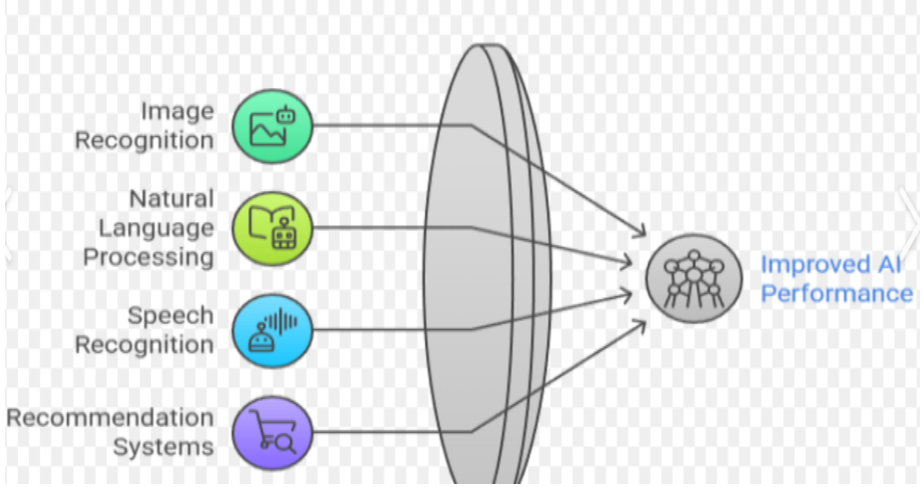
- **Architecture Design:**

1. Choosing number of layers, neurons, activation functions.
2. Good design avoids overfitting and increases accuracy.
3. Essential for deep learning models.



10. What is Gradient-Based Learning?

- Uses gradient (slope) to reduce error.
- Loss function gives direction to update weights.
- Works with gradient descent and its variants.
- Core method for training neural networks.



- **Purpose**: Optimize model parameters to **minimize a loss function**.
- **Idea**: Use the **gradient (slope) of the loss function** to guide updates to parameters.
- **Key Concept**: Move parameters in the **opposite direction of the gradient** to reduce error.
- **Steps**:
  - Initialize parameters randomly.
  - Compute gradient of loss w.r.t parameters.
  - Update parameters: $\theta = \theta - \text{learning_rate} \times \text{gradient}$
  - Repeat until loss is minimized.
- **Example**: Training neural networks using backpropagation and gradient descent.
- **Advantage**: Efficient for large and complex models like deep learning.

11. What is Backpropagation? Explain with a neat diagram.

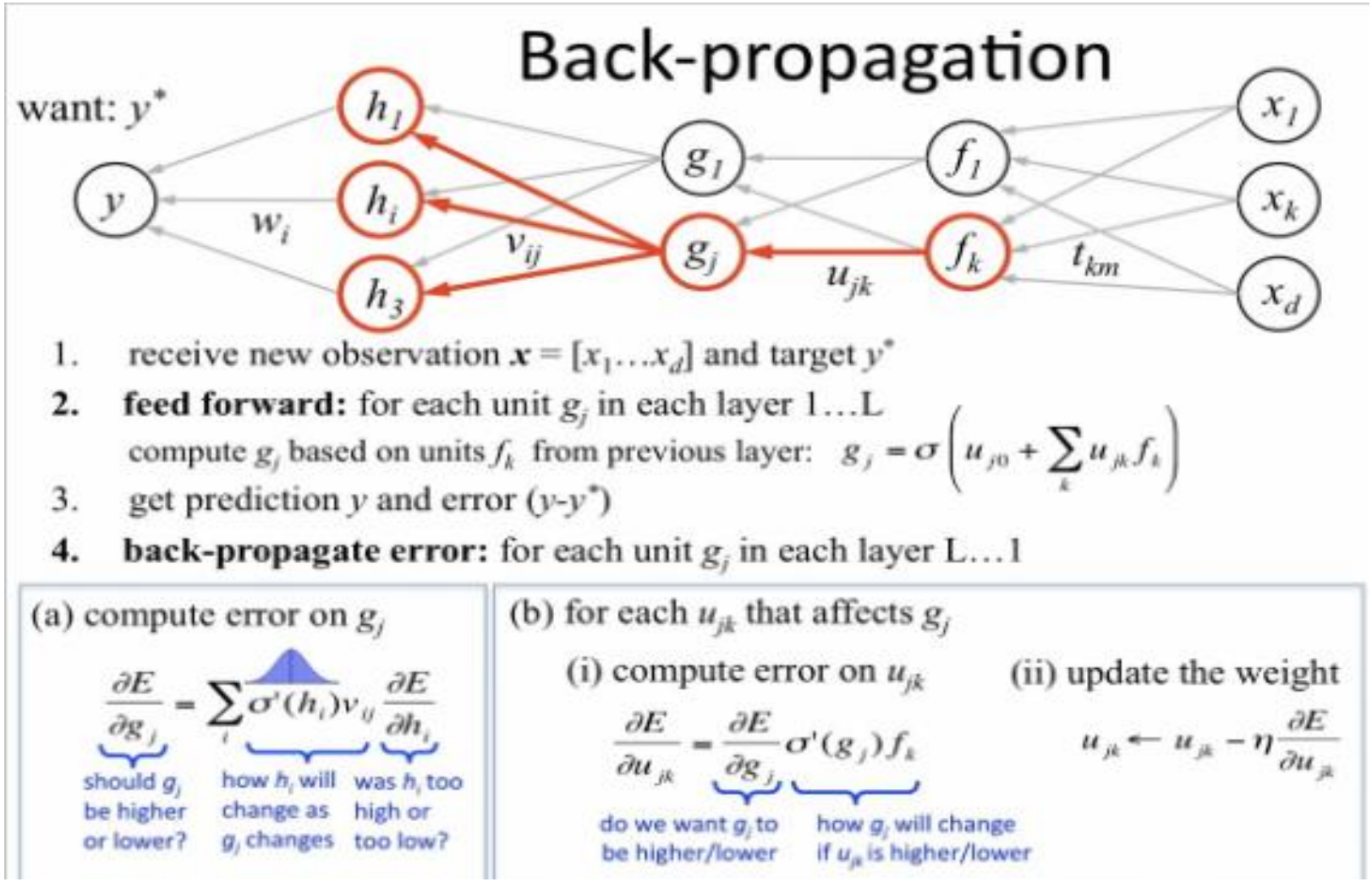
- **Back-Propagation:**

1. Calculates error at output.
2. Sends error backward through layers.
3. Adjusts weights to reduce error.
4. Works with gradient descent.

- **Other Algorithms:**

1. Momentum – speeds up learning.
2. RMSProp – stabilizes updates.
3. Adam – fast, efficient, widely used in deep learning.

Back Propagation



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